Motor Current Histogram

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Relevant Links:

- N/A (this was a one-off request)

Imports

Import any packages or dependencies for this project. Uncomment or delete these lines as needed, and add other dependencies

In [24]:
```
from datetime import timedelta
import numpy as np
import matplotlib.pyplot as plt
from data_tools import query, FSGPDayLaps, TimeSeries
```

Data Acquisition

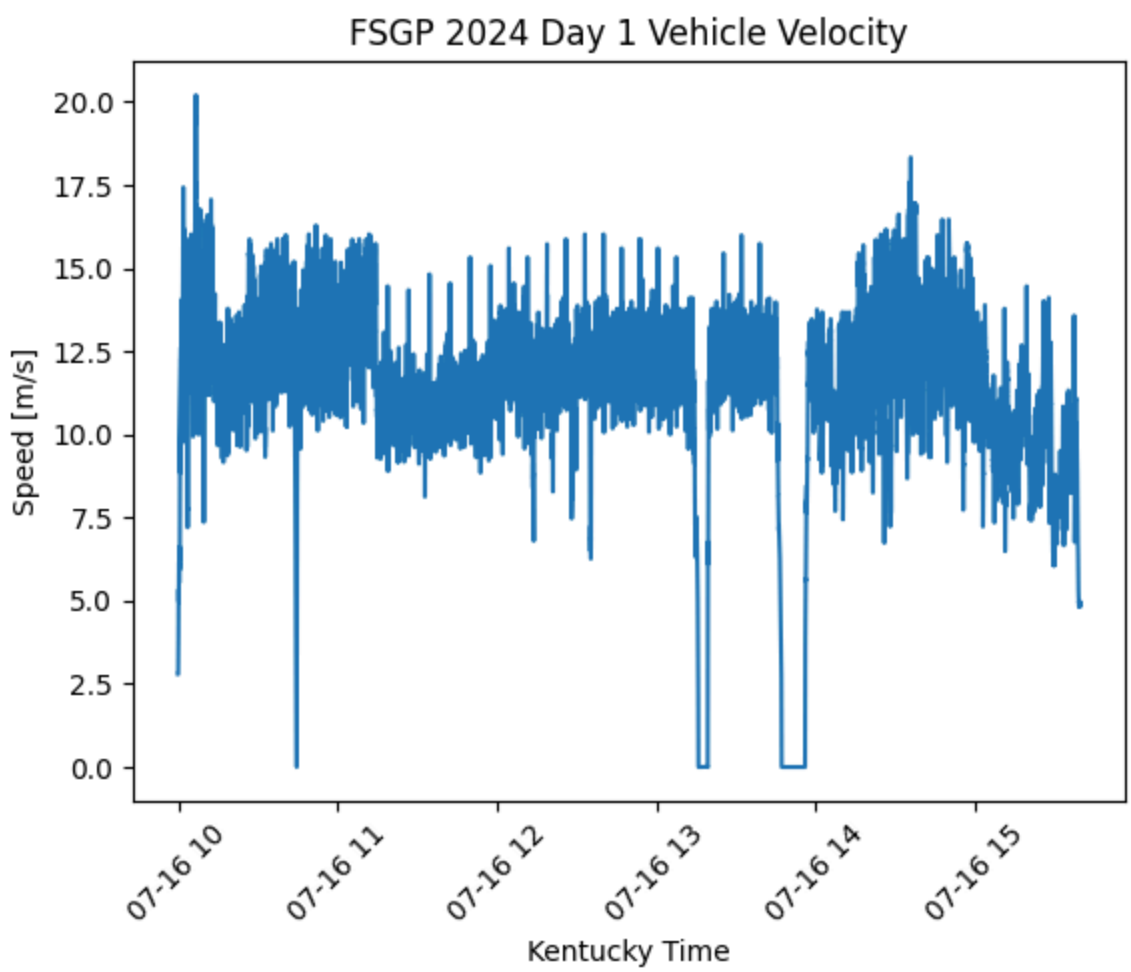
Obtain the data relevant to this analysis. Load data from saved files, or query it directly.

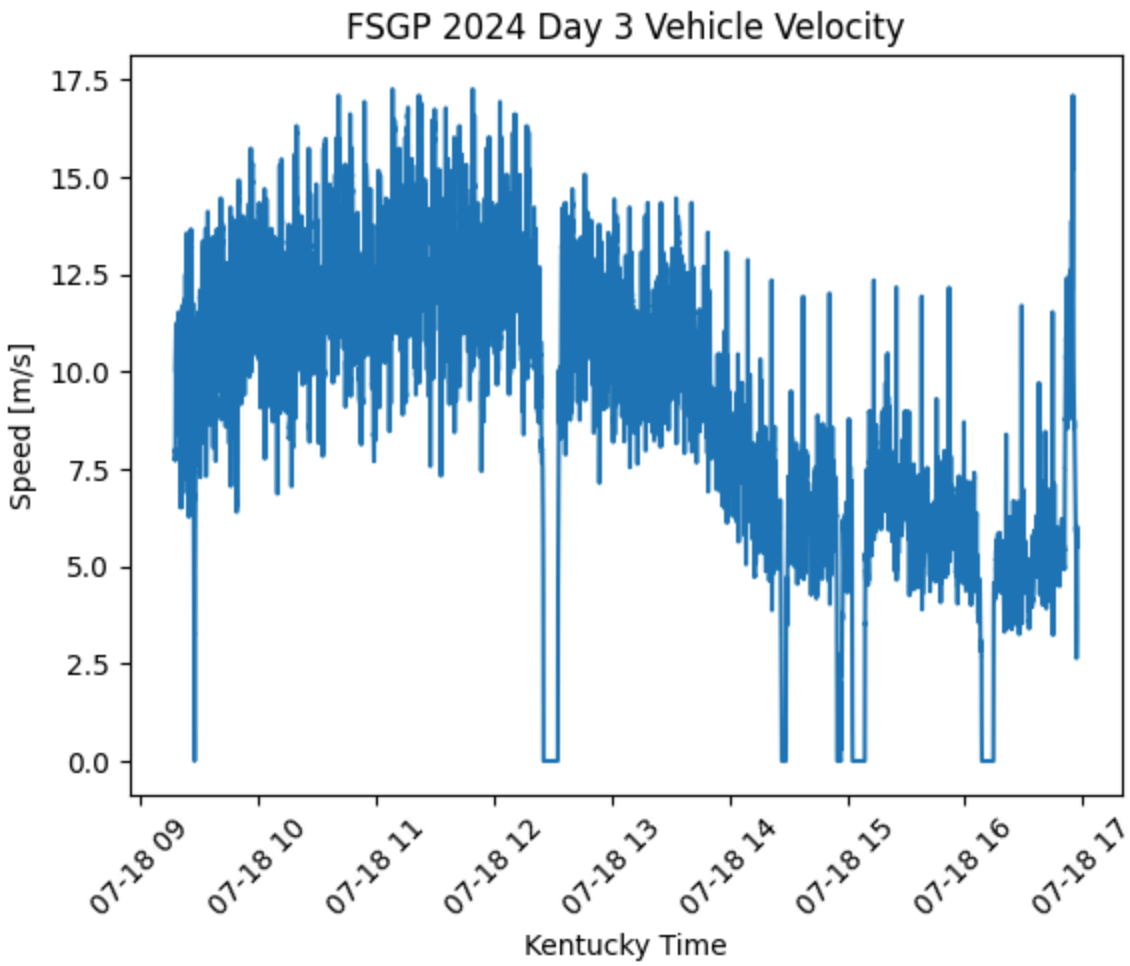
In [19]:
```
client = query.DBClient()

fields = ('BatteryCurrent', 'VehicleVelocity')
data: dict[int, dict[str, TimeSeries]] = {}

for day in (1, 3):
    race_day = FSGPDayLaps(day)
    start = race_day.get_start_utc(1)
    last_lap = race_day.get_lap_count()
    stop = race_day.get_finish_utc(last_lap)

    day_data = {}
    for field in fields:
        day_data[field] = client.query_time_series(start, stop, field)

    data[day] = day_data
```

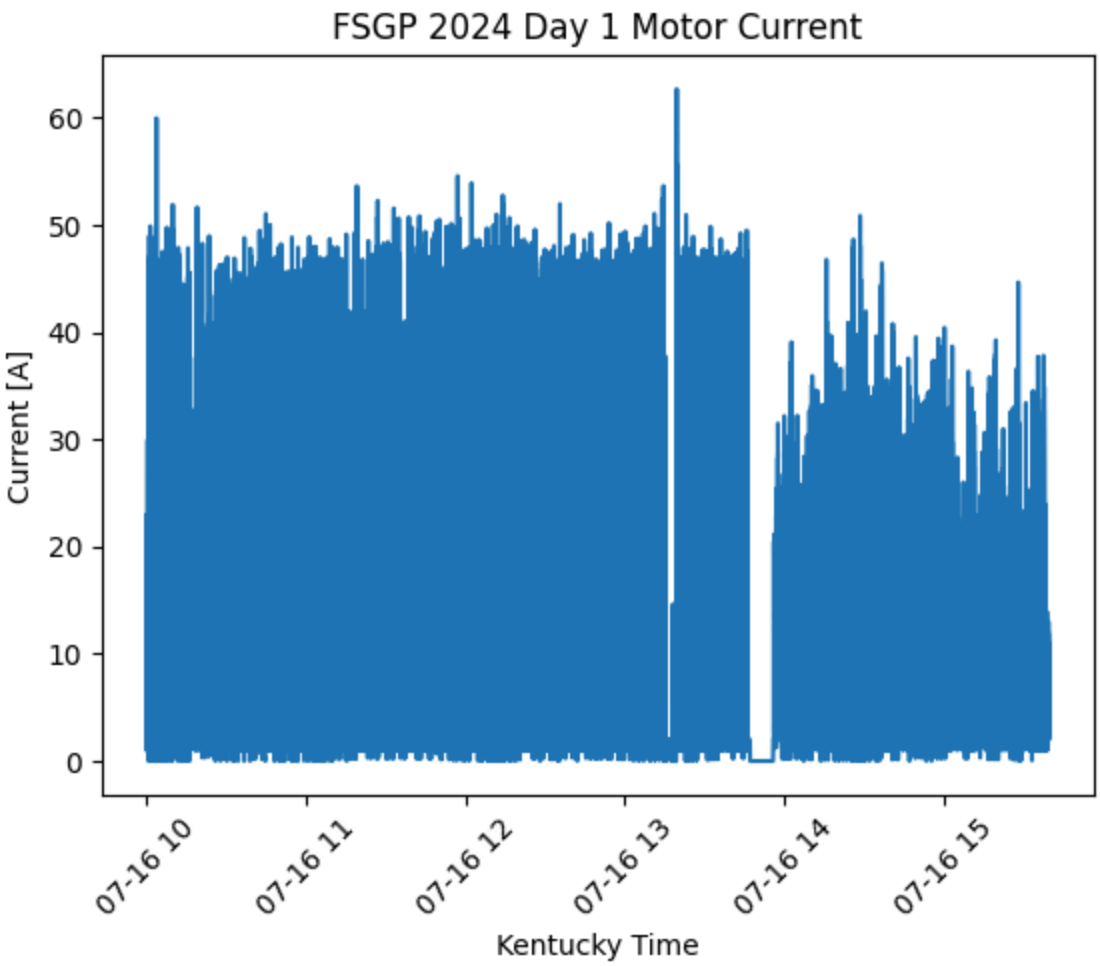
Analysis

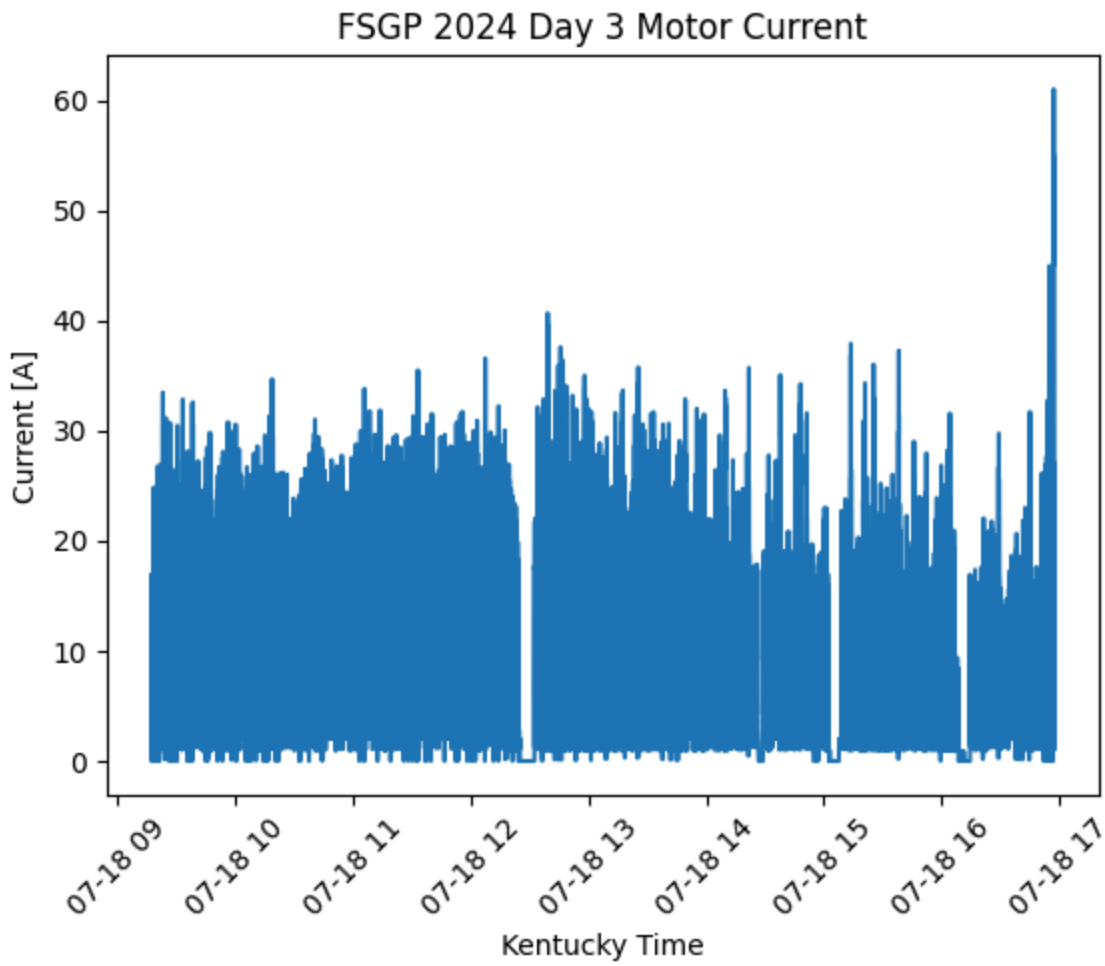
Compute the desired results and present them. Take advantage of Markdown cells to guide the reader through your derivation process.

In [25]:
```
vancouver_to_kentucky_time: timedelta = timedelta(hours=2)

for day, day_data in data.items():
    speed_mps = day_data['VehicleVelocity']
    plt.plot(speed_mps.datetime_x_axis + vancouver_to_kentucky_time, speed_mps)
    plt.title(f"FSGP 2024 Day {day} Vehicle Velocity")
    plt.xlabel("Kentucky Time")
    plt.ylabel("Speed [m/s]")
    plt.xticks(rotation=45)
    plt.show()

for day, day_data in data.items():
    current = day_data['BatteryCurrent']
    plt.plot(current.datetime_x_axis + vancouver_to_kentucky_time, current)
    plt.title(f"FSGP 2024 Day {day} Motor Current")
    plt.xlabel("Kentucky Time")
    plt.ylabel("Current [A]")
    plt.xticks(rotation=45)
    plt.show()
```

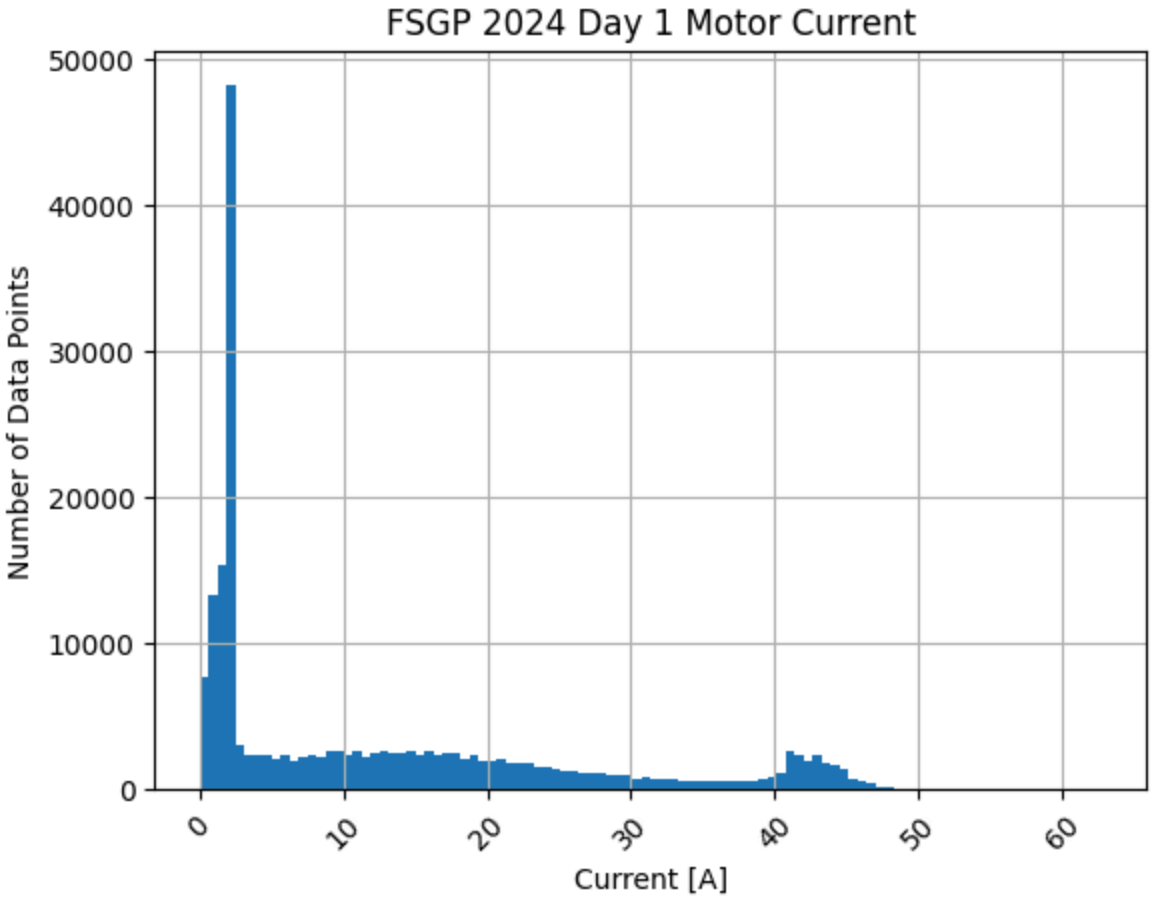


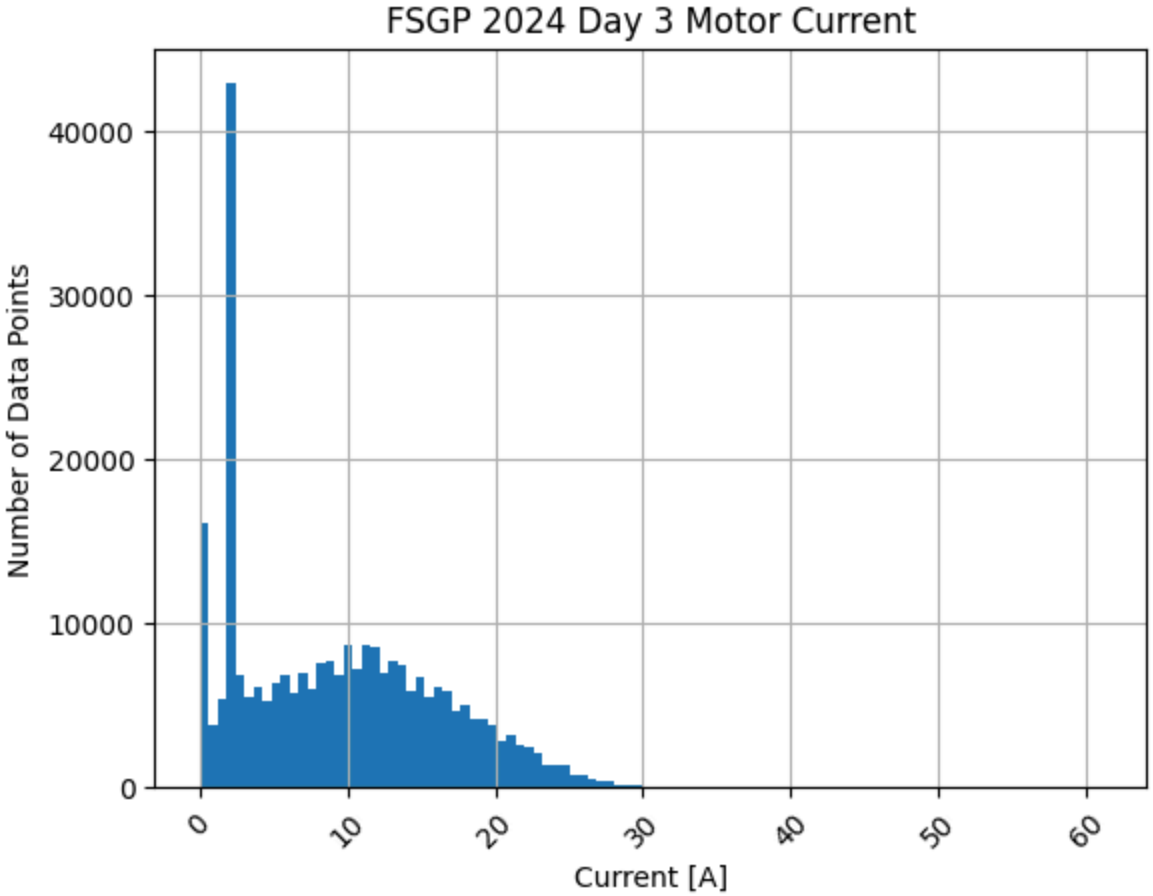

FSGP 2024 Day 3 Vehicle Velocity
Speed [m/s]
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
07-18 09
07-18 10
07-18 11
07-18 12
07-18 13
07-18 14
07-18 15
07-18 16
07-18 17
Kentucky Time


FSGP 2024 Day 1 Motor Current
Current [A]
60
50
40
30
20
10
0
07-16 10
07-16 11
07-16 12
07-16 13
07-16 14
07-16 15
Kentucky Time



In [26]:
```
for day, day_data in data.items():
    current = day_data['BatteryCurrent']
    plt.hist(current, bins=100)
    plt.title(f"FSGP 2024 Day {day} Motor Current")
    plt.xlabel("Current [A]")
    plt.ylabel("Number of Data Points")
    plt.xticks(rotation=45)
    plt.grid()
    plt.show()
```

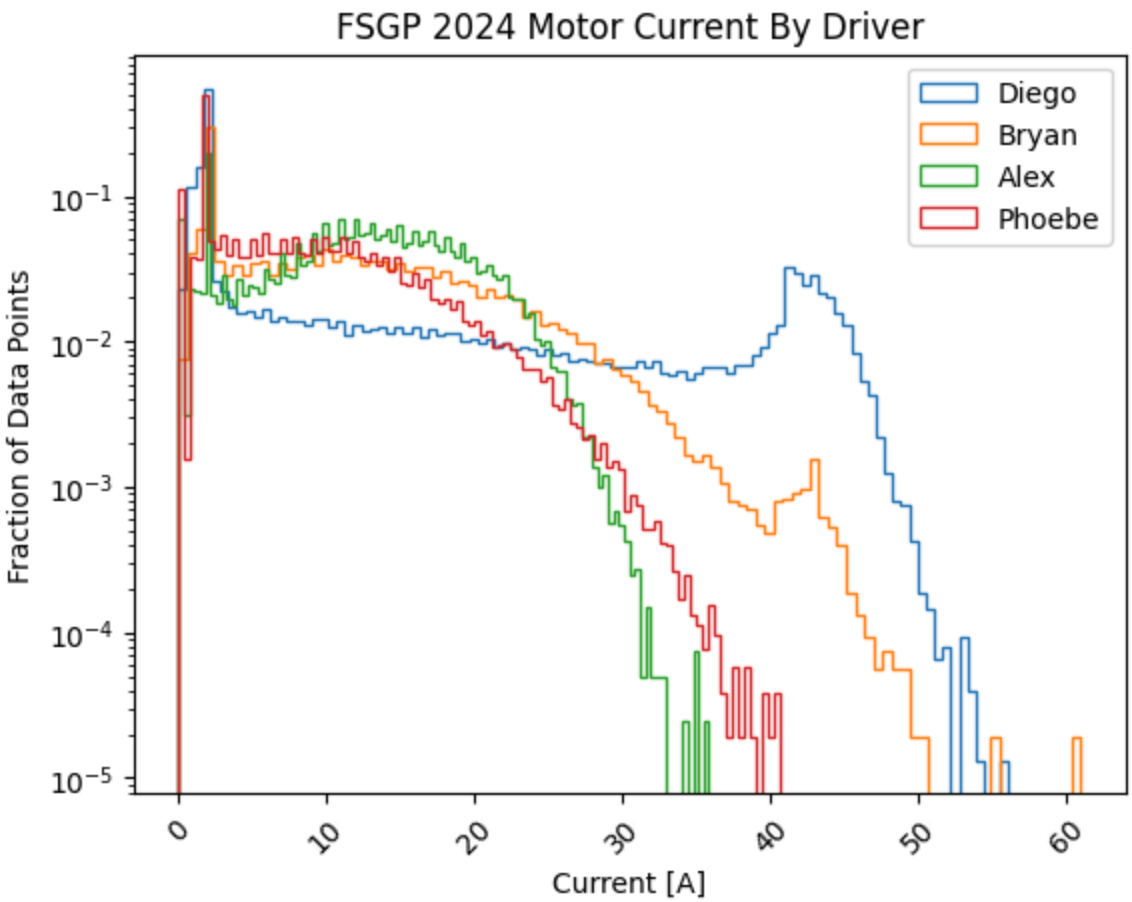

FSGP 2024 Day 1 Motor Current
Number of Data Points
Current [A]
0
10000
20000
30000
40000
50000
0
10
20
30
40
50
60


FSGP 2024 Day 3 Motor Current
Number of Data Points
Current [A]
0
10000
20000
30000
40000
0
10
20
30
40
50
60

In [33]:
```
driver_currents: dict[str, list[TimeSeries]] = {}

for day in (1, 3):
    race_day = FSGPDayLaps(day)

    for lap_idx in range(race_day.get_lap_count()):
        lap = lap_idx + 1
        driver = race_day.get_lap_driver(lap)

        start = race_day.get_start_utc(lap)
        stop = race_day.get_finish_utc(lap)
        if driver not in driver_currents:
            driver_currents[driver] = []
        driver_currents[driver].append(
            client.query_time_series(start, stop, 'BatteryCurrent')
        )

        data[day] = day_data
```

In [34]:
```
driver_currents_joined = {}
for driver, laps in driver_currents.items():
    driver_currents_joined[driver] = np.concatenate([lap_data for lap_data in laps]
```

In [44]:
```
for driver, all_current_data in driver_currents_joined.items():
    plt.hist(all_current_data, bins=100, histtype='step', label=driver, density=Tru
    plt.title(f"FSGP 2024 Motor Current By Driver")
    plt.xlabel("Current [A]")
    plt.ylabel("Fraction of Data Points")
    plt.xticks(rotation=45)
    plt.grid()
plt.legend(loc='best')
plt.yscale('log')
plt.show()
```


FSGP 2024 Motor Current By Driver
Diego
Bryan
Alex
Phoebe
Fraction of Data Points
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
10
20
30
40
50
60
Current [A]

In [ ]: